NAVEEN PERERA, COMPUTER SCIENCE UNDERGRADUATE

Kadawatha, Western Province 11850, Sri Lanka, +94 78 870 0322, naveenmarioperera04@gmail.com

LINKS LinkedIn GitHub Portfolio

TECHNICAL SKILLS

Java	Python
SQL	Dart
JavaScript	HTML5/CSS3
Node.js	Express
React	Redux Toolkit
Tailwind CSS	PostgreSQL
Docker	Nginx
OAuth 2.0 / OpenID Connect (OIDC)	JWT, WebAuthn (FIDO2)
Flutter	Tkinter
Pandas	NumPy
Scikit-learn	Plotly Express
GridSearchCV	MySQL
Firebase (Firestore, Auth, Storage, Hosting)	Figma
Graphic Design	UI/UX Planning
Photography & Videography	Video Editing
Git	GitHub
Google Colab	Loom

PROJECTS

Jul 2026 — Nov 2026

Identity Platform (OAuth 2.0 & OIDC SSO) — SLT Internship

Node.js, Express, React, PostgreSQL, WebAuthn, Docker

- Developed a full-stack OAuth 2.0 / OpenID Connect identity provider with Authorization Code Flow + PKCE, signed JWTs and JWKS discovery, and refresh token rotation.
- Integrated passwordless FIDO2/WebAuthn passkeys and Email/SMS OTP verification to enable single sign-on across multiple applications.
- Designed a sign-in activity and security logging pipeline with automated risk scoring and suspicious-login email alerts, aligned with data protection requirements.
- Built a React, Redux Toolkit and Tailwind CSS admin console with role-based access control, analytics dashboards and light/dark theming.
- Containerized the platform with Docker and Nginx, and applied security hardening including rate limiting and vulnerability scanning.

Aug 2026 — Sep 2026

ZenData — Automated Data Quality & Semantic Hygiene Platform

Python, FastAPI, Pandas, NumPy, Next.js, TypeScript, ReportLab, OpenPyXL | Live Demo | GitHub

- Engineered a 4-Pillar Data Health Audit Engine evaluating tabular datasets across Completeness, Uniqueness, Validity, and Statistical Consistency (using 1.5× IQR outlier detection) to generate real-time 0–100% quality scores.
- Implemented Semantic Hygiene & Anti-Poisoning Rules in Pandas to purge sentinel placeholders (?, not given) and cluster categorical typos (Diary → Dairy) without fabricating values for missing PII or transaction IDs.
- Designed Mathematical Reconciliation Logic cross-validating financial formulas (Total = Qty × Price × (1 - Discount)) to reverse-engineer corrupted totals, outlier quantities, and inverted unit prices.
- Automated Executive Reporting Pipelines leveraging ReportLab and OpenPyXL to generate landscape PDF quality audit scorecards and 3-sheet styled Excel workbooks with full transformation changelogs.

May 2026

Loan Approval Prediction System
Machine Learning & Data Mining
Sole Developer

- Built an end-to-end ML classification pipeline in Python (Pandas, NumPy, Scikit-learn, Plotly) to predict loan approvals on a 58,645-record financial dataset.
- Performed EDA and cleaned the data — missing value imputation, outlier capping via IQR, feature encoding, and scaling — to engineer a model-ready dataset.
- Trained Logistic Regression, Gaussian Naive Bayes, and KNN classifiers with a stratified train-test split, achieving up to 99.97% accuracy and an AUC-ROC of 1.0.
- Optimised model performance using GridSearchCV with 5-fold cross-validation across multiple hyperparameter combinations.

May 2026

Suspicious Log Analyzer — Security Mini-Project
Sole Developer

- Developed a Python-based incident response tool to monitor server access logs and detect unauthorized access attempts.
- Utilized data mining techniques and dictionary structures to track IP behavior and authentication statuses in real time.
- Implemented a threat-detection threshold to automatically flag potential brute-force attacks for firewall blocking.

Mar 2025 — Jun 2025

Civic Engagement and Feedback Platform — SpotIT LK
Backend & Multimedia Integration Lead

- Architected and implemented the multimedia storage infrastructure for a civic engagement mobile app, enabling citizens to upload and attach photo evidence to public issue reports.
- Integrated Firebase Cloud Storage to securely host, manage, and retrieve user-generated media, ensuring scalable and efficient data delivery to government dashboards.
- Developed seamless image capture and upload workflows within the Flutter frontend.
- Implemented image compression techniques to minimize data usage for users on low-bandwidth networks.
- Collaborated on the overall client-server architecture, ensuring stable synchronization between the mobile app, Firebase Firestore, and cloud storage services.
- Supported SDG Goal 11 by building robust digital infrastructure for transparent, community-driven governance solutions.

EDUCATION

2025 — Present	BSc Computer Science, University of Westminster	UK (at IIT)
2023	G.C.E. Advanced Level, De Mazenod College Physical Science Stream	Kandana
	G.C.E. Ordinary Level, De Mazenod College	Kandana

CERTIFICATIONS

Professional Certificate in Cyber Security & Networking, IIT School of Computing
Certificate in Networking and System Administration
Certificate in Cyber Security Information

LANGUAGES

English Sinhala

INTERESTS

Photography & Videography, Badminton, Tennis, Cricket

REFERENCES

Krishan Tharaka, Senior Technical Configuration Specialist from Arcadea Group
krishan.tharake@gmail.com · 077 848 3300

Pahan Gannile, Executive – Digital Marketing & IT
pinsarapahan1@gmail.com · 070 300 3822

LEADERSHIP & VOLUNTEER

- Project Leadership: Led teams to successful technical outcomes and collaborated on complex software development projects with multiple stakeholders.
- Athletics & Discipline: Represented De Mazenod College in Badminton, Tennis, and Cricket, building strong collaborative skills, teamwork, and focus through competitive sports.
- Core Soft Skills: Honed critical thinking, creativity, and effective problem-solving abilities through a balance of rigorous academic work and extracurricular commitments.

KEY HIGHLIGHTS

- SDG 11 Aligned Project (SpotIT LK)
- 99.97% Accuracy — Loan Approval ML Model
- Team Leadership
- Full-Stack Development